Lecture 7

① Theorem : $\varepsilon_{ij}, \varepsilon_{i\alpha}, \varepsilon_{ij\alpha} \in Aut_K(V^m)$

$f: V \to V'$ is a K - linear map / vector space homom

if: $\begin{cases} f(v_1 + v_2) = f(v_1) + f(v_2), & (\forall) v_1, v_2 \in V \\ f(k v) = k f(v), & (\forall) k \in K, (\forall) v \in V. \end{cases}$

$\left.\begin{array}{l} f \to K \text{ linear map} \\ V = V' \\ f \text{ bij.} \end{array}\right\} \Rightarrow f$ automorphism.

Proof: $\varepsilon_{ij}, \varepsilon_{i\alpha}, \varepsilon_{ij\alpha} : V^m \to V^m$.

- Show that V^m has a structure of a vector space.

$(v_1, \dots, v_m) + (v_1', \dots, v_m') = (v_1 + v_1', \dots, v_m + v_m')$

$k(v_1, \dots, v_m) = (k v_1, \dots, k v_m)$

- Check that $\varepsilon_{ij}, \varepsilon_{i\alpha}, \varepsilon_{ij\alpha}$ are K-linear maps

$\varepsilon_{ij}(v_1, \dots, v_i, \dots, v_j, \dots, v_m) = (v_1, \dots, v_j, \dots, v_i, \dots v_m)$

$\varepsilon_{i\alpha}(v_1, \dots, v_i, \dots, v_m) = (v_1, \dots, \alpha v_i, \dots, v_m)$

$\varepsilon_{ij\alpha}(v_1, \dots, v_i, \dots, v_j, \dots, v_m) =$

$= (v_1, \dots, v_i + \alpha v_j, \dots, v_j, \dots, v_m)$

- + bijections because they have inverses:

$(\varepsilon_{ij})^{-1} = \varepsilon_{ji}, (\varepsilon_{i\alpha})^{-1} = \varepsilon_{i\alpha^{-1}}, (\varepsilon_{ij\alpha})^{-1} = \varepsilon_{ij(-\alpha)}$

$\Rightarrow$ linear maps + bij + $V^m = V^m \Rightarrow$ automorphisms.

② Theorem: ${}_K V$, X and X' equiv. lists of vectors.

1) X lin. ind. in $V^m \Leftrightarrow X'$ lin. ind. in V^m

2) X system of generators $\Leftrightarrow X'$ sys. of gen.

3) X basis of $V^m \Leftrightarrow X'$ basis of V^m.

Proof

- We know that X, X' equiv. lists $\Rightarrow$

  $\exists\, \varphi: V^m \to V^m$ s.t. $\varphi(X) = X'$

  $\left.\begin{array}{l}\varphi \to \text{composition of elementary operations}\\ \text{each elementary operation} \to \text{isomorphism}\end{array}\right\} \Rightarrow$

  $\Rightarrow \boxed{\varphi \text{ isomorphism}}$

- isomorphisms preserve
  - lin. ind. lists
  - systems of gen.
  - bases

③. **Theorem:** The value of an elementary operation applied on a matrix $A = (a_{ij}) \in \mathcal{M}_{m,n}(K)$, seen as a list of column-vectors $(a^1, \dots, a^n)$ is equal to A multiplied on the right hand side by the matrix obtained from I_n, also seen as a list of columns, by applying the same elem. operation.

Proof
(for the first two columns)

E_{ij}

$$E_{12}(A) = E_{12}(a^1, a^2, \dots, a^n) = (a^2, a^1, \dots, a^n)$$

$$= \begin{pmatrix} a_{12} & a_{11} & \dots & a_{1n} \\ a_{22} & a_{21} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m2} & a_{m1} & & a_{mn} \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} 0 & 1 & \dots & 0 \\ 1 & 0 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & 1 \end{pmatrix}}_{E_{12}}$$

E_{12} = the matrix obtained from I_n by <u>interchanging</u> the first two columns.

$\mathcal{E}_{i\alpha}$

$$\mathcal{E}_{1\alpha}(A) = \mathcal{E}_{1\alpha}(a^1, a^2, \dots, a^m) = (\alpha a^1, a^2, \dots, a^m)$$

$$= \begin{pmatrix} \alpha a_{11} & a_{12} \dots & a_{1m} \\ \vdots & \vdots & \vdots \\ \alpha a_{m1} & a_{m2} & a_{mm} \end{pmatrix}$$

$$= \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mm} \end{pmatrix} \cdot \begin{pmatrix} \alpha & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = A \cdot E_{1\alpha}$$

$E_{1\alpha}$ = the matrix obtained from I_m by <u>multiplying</u> the first column by α.

$\mathcal{E}_{ij\alpha}$

$$\mathcal{E}_{12\alpha}(A) = \mathcal{E}_{12\alpha}(a^1, a^2, \dots, a^m) = (a^1 + \alpha a^2, \dots, a^m)$$

$$= \begin{pmatrix} a_{11} + \alpha_{12} & a_{12} & \dots & a_{1m} \\ a_{21} + \alpha_{22} & a_{22} & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{m1} + \alpha a_{m2} & a_{m2} & \dots & a_{mm} \end{pmatrix}$$

$$= \underbrace{\begin{pmatrix} a_{11} & \dots & a_{1m} \\ a_{21} & & \\ \vdots & & \\ a_{m1} & \dots & a_{mm} \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} 1 & 0 & \dots & 0 \\ \alpha & 1 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & & & \\ 0 & 0 & \dots & 1 \end{pmatrix}}_{E_{12\alpha}}$$

$E_{12\alpha}$ = the matrix obtained ~~by~~ from I_m by <u>multiplying</u> the second column by α and <u>adding</u> it to the first column.

4. Theorem: Elementary matrices (E_{ij}, $E_{i\alpha}$, $E_{ij\alpha}$) are INVERTIBLE.

Proof:

$\det I_n = 1 \neq 0$

The determinant remains non-zero after applying elementary operations on I_n.

$\Rightarrow$ those matrices (E_{ij}, $E_{i\alpha}$, $E_{ij\alpha}$) invertible

5. Theorem: Every non-zero matrix is equivalent to a matrix in echelon form.

Proof: (how to get to echelon)

Let $A = (a_{ij}) \in \mathcal{M}_{m,n}(K)$.

- Look for a row with the least zeroes from the beginning of the row
- Interchange it with the first row.
- Pivot = a_{1j_1} = the first $\neq 0$ element from row 1.
- Make zeroes on column j_1 $\rightarrow$ elem. operations on rows
- Look for a row with the least zeroes from the beginning of the row
- Interchange it with the second row.
- Pivot = a_{2j_2} = the first $\neq 0$ elem. from row 2.
- Make zeroes on column j_2 $\rightarrow$ elem. operations...)

⑥. **Lemma**: $A = (a_{ij}) \in \mathcal{M}_{m,n}(K)$ seen as a list of row-vectors $(a_1, \dots, a_m)$. Then a sublist of $(a_1, \dots, a_m)$ consisting of r vectors is lin. indep. in K^n $\Leftrightarrow$ $\exists$ a non-zero minor of order r of the matrix A.

Proof.

$X = (a_1, \dots, a_r)$ sublist of $(a_1, \dots, a_m)$.

X lin. indep. $\Leftrightarrow$ $k_1 a_1 + \dots + k_r a_r = 0$.

$\Rightarrow k_1 = k_2 = \dots = k_r = 0$ $\Leftrightarrow$

$k_1, \dots, k_r \in K$.

$$\Leftrightarrow \begin{cases} k_1 a_{11} + k_2 a_{21} + \dots + k_r a_{r1} = 0. \\ k_1 a_{12} + k_2 a_{22} + \dots + k_r a_{r2} = 0 \\ \vdots \\ k_1 a_{1n} + k_2 a_{2n} + \dots + k_r a_{rn} = 0. \end{cases}$$

$\Leftrightarrow$ $\exists$ a non-zero minor of order r of A.

(7) **Theorem:** Let $0 \neq A = (a_{ij}) \in M_{m,n}(K)$, seen as a list of row-vectors $(a_1, \dots, a_m)$ or as a list of column-vectors $(a^1, \dots, a^n)$. Then:

1) $\text{rank}(A) = \dim \langle a_1, \dots, a_m \rangle = \dim \langle a^1, \dots, a^n \rangle$

2) $\text{rank}(A) = \text{rank}(C) = r$

C = matrix in echelon form with $r \neq 0$ rows $\sim A$.

Proof.

1) $r = \dim \langle a_1, \dots, a_m \rangle$ (row)

max. nr. of lin. ind. vectors in $(a_1, \dots, a_m)$ is r

max. order of $\neq 0$ minors of A is r.

$\Rightarrow \text{rank}(A) = r$.

$r = \dim \langle a^1, \dots, a^n \rangle$ (column)

$\Rightarrow \text{rank}(A) = r$.

2) $A \to$ list of row-vectors $(a_1, \dots, a_m)$

$C \to$ list of row-vectors $(c_1, \dots, c_r, c_{r+1}, \dots c_m)$

$\Rightarrow c_{r+1} = \dots = c_m = 0$ using Lemma

$\Rightarrow (c_1, \dots, c_r)$ lin. indep. list.

! Elementary operations preserve the rank.

$$\begin{aligned} \Rightarrow \text{rank}(A) &= \dim \langle a_1, \dots, a_m \rangle \\ &= \dim \langle c_1, \dots, c_m \rangle \\ &= \dim \langle c_1, \dots, c_r \rangle = r = \text{rank}(C) \end{aligned}$$

8 **Theorem**: Let $A \in \mathcal{M}_n(K)$ with $\det(A) \neq 0$. $\Rightarrow$

$\Rightarrow$ $A \sim I_n$

$\Rightarrow$ A^{-1} is obtained from I_n by applying the same elementary operations as one does to obtain I_n from A.

Proof:

- $\det(A) \neq 0 \Rightarrow A$ invertible. $\Rightarrow \operatorname{rank}(A) = n$.
- elementary operations $\Rightarrow$ C in echelon form.
  (n non-zero rows)
  (triangular matrix)
- make zeroes above the principal diagonal by applying elem. operations on ROWS starting from the last row.
- $A \sim$ a matrix in diagonal form. (triangular)
- $\operatorname{rank}(A) = n \Rightarrow$ all the elem. on the diagonal $\neq 0$

$\Rightarrow$

$\Rightarrow$ multiply by their inverses $\Rightarrow I_n$.

$\Rightarrow A \sim I_n$.

?

$\Rightarrow \underbrace{\exists\, E_1, \ldots, E_K}_{\text{elementary matrices}}$ s.t. $E_K \ldots E_1 A = I_n$.

$\Rightarrow A^{-1} = E_K \ldots E_1 I_n$

A^{-1} obtained from I_n by applying the same elementary operations as $A \to I_n$.

Lecture 8

① **Theorem**: Let $f : V \to V'$ a K - linear map.

$B = (v_1, \dots, v_n)$ basis of V

$B' = (v_1', \dots, v_m')$ basis of V'

$v \in V$

Then $\boxed{[f(v)]_{B'} = [f]_{BB'} \cdot [v]_B}$

Proof:

- $[f]_{BB'} = (a_{ij}) \in \mathcal{M}_{m,n}(K)$
- $v = \sum_{j=1}^{n} k_j v_j \;/ f \Rightarrow f\left(\sum_{j=1}^{n} k_j v_j\right) = \sum_{j=1}^{n} k_j \cdot \underbrace{f(v_j)}_{(*)}$
- $f(v) = \sum_{i=1}^{m} k_i' v_i' \quad (**)$

?

$\Rightarrow f(v) = \sum_{j=1}^{n} k_j \left(\sum_{i=1}^{m} a_{ij} \cdot v_i'\right)$

$= \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \cdot k_j\right) \cdot v_i' = (**)$

$\Rightarrow$ because of uniqueness, we must have:

$\sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \cdot k_j\right) \cdot v_i' = \sum_{i=1}^{m} k_i' \cdot v_i'$

$\Rightarrow k_i' = \sum_{j=1}^{n} a_{ij} \cdot k_j, \; (\forall)\, i = \overline{1,m}$

$\Rightarrow \cancel{[}f(v)\cancel{]_{B'}} = \sum_{i=1}^{m} \left(\sum_{j=1}^{n} a_{ij} \cdot k_j\right) \cdot v_i'$

$$[f(v)]_{B'} = \begin{pmatrix} \sum_{j=1}^{m} a_{1j} \cdot k_j \\ \vdots \\ \sum_{j=1}^{m} a_{mj} \cdot k_j \end{pmatrix} \quad (1)$$

$$[f]_{BB'} = \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & & \\ a_{m1} & \cdots & a_{mm} \end{pmatrix}$$

$$[v]_B = \begin{pmatrix} k_1 \\ \vdots \\ k_m \end{pmatrix}$$

$$\Rightarrow [f]_{BB'} \cdot [v]_B = \begin{pmatrix} a_{11} \cdot k_1 + a_{12} \cdot k_1 + \cdots + a_{1m} \cdot k_1 \\ \vdots \\ a_{m1} \cdot k_m + \cdots + a_{mm} \cdot k_m \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{j=1}^{m} a_{1j} \cdot k_j \\ \vdots \\ \sum_{j=1}^{m} a_{mj} \cdot k_j \end{pmatrix} \quad (2)$$

$\Rightarrow (1) = (2)$.

$$\Rightarrow [f(v)]_{B'} = [f]_{BB'} \cdot [v]_B.$$

② **Theorem** : Let $f : V \to V'$ linear map.

$\operatorname{rank}(f) = \cancel{\operatorname{rank}(BB')}\ \operatorname{rank}([f]_{BB'})$

$B, B' \to$ any bases of V, V'.

Proof.

$B = (v_1, \dots, v_m)$

$[f]_{BB'} = A$

$$\operatorname{rank}(f) = \dim(\operatorname{Im} f) = \dim(f(V))$$

?

$$= \dim f(\langle v_1, \dots, v_m \rangle)$$

lin. map

$$= \dim(\langle f(v_1), \dots, f(v_m) \rangle)$$

?

$$= \operatorname{rank}(A^T)$$

$$= \operatorname{rank}(A)$$

$$= \operatorname{rank}([f]_{BB'})$$

$B_1 = (u_1, \dots, u_m)$ basis of V.

$B_1' \Rightarrow$ basis of V'.

$[f]_{B_1B_1'} = A_1$.

$$\operatorname{rank}([f]_{B_1B_1'}) = \operatorname{rank}(A_1) = \operatorname{rank}(A_1^T)$$

?

$$= \dim \langle f(u_1), \dots, f(u_m) \rangle = \dim(\operatorname{Im} f)$$

$$= \dim \langle f(v_1), \dots, f(v_m) \rangle = \operatorname{rank}([f]_{BB'})$$

③ <u>Theorem</u> : V, V', V'' vector spaces over K.

$\dim V = n$, $\dim V' = m$, $\dim V'' = p$.

$B = (v_1, \dots, v_n)$, $B' = (v_1', \dots, v_m')$, $B'' = (v_1'', \dots, v_p'')$

$(\forall) f, g \in \mathrm{Hom}_K(V, V')$

$(\forall) h \in \mathrm{Hom}_K(V', V'')$

$(\forall) k \in K$

$\Rightarrow [f+g]_{BB'} = [f]_{BB'} + [g]_{BB'}$

$\Rightarrow [kf]_{BB'} = k \cdot [f]_{BB'}$

$\Rightarrow [h \circ f]_{BB''} = [h]_{B'B''} \cdot [f]_{BB'}$.

<u>Proof</u> :

$[f]_{BB'} = (a_{ij}) \in \mathcal{M}_{m,n}(K)$

$[g]_{BB'} = (b_{ij}) \in \mathcal{M}_{m,n}(K)$

$[h]_{B'B''} = (c_{ki}) \in \mathcal{M}_{m,n}(K)$?

$$\Rightarrow f(v_j) = \sum_{i=1}^{m} a_{ij} \cdot v_i'$$

$$\Rightarrow g(v_j) = \sum_{i=1}^{m} b_{ij} \cdot v_i'$$

$$\Rightarrow h(v_i') = \sum_{k=1}^{p} c_{ki} \cdot v_k''$$

$$(f+g)(v_j) = f(v_j) + g(v_j) = \sum_{i=1}^{m} a_{ij} v_i' + \sum_{i=1}^{m} b_{ij} \cdot v_i'$$

$$= \sum_{i=1}^{m} (a_{ij} + b_{ij}) \cdot v_i'$$

$$(kf)(v_j) = k f(v_j) = k \cdot \left(\sum_{i=1}^{m} a_{ij} \cdot v_i' \right) = \sum_{i=1}^{m} (k \cdot a_{ij}) \cdot v_i'$$

?

$\Rightarrow [f+g]_{BB'} = [f]_{BB'} + [g]_{BB'}$

$\Rightarrow [kf]_{BB'} = k \cdot [f]_{BB'}$.

$$(h \circ f)(v_j) = h(f(v_j)) = h\left(\sum_{i=1}^{m} a_{ij} \cdot v_i' \right) =$$

$$= \sum_{i=1}^{m} a_{ij} \cdot \underbrace{h(v_i')}$$

$$= \sum_{i=1}^{m} a_{ij} \cdot \left(\sum_{k=1}^{p} c_{ki} \cdot v_k'' \right)$$

$$= \sum_{k=1}^{p} \sum_{i=1}^{m} (c_{ki} \cdot a_{ij}) \cdot v_k''$$

?

$\Rightarrow [h \circ f]_{BB''} = [h]_{B'B''} \cdot [f]_{BB'}$.

4. <u>Theorem</u>: ${}_K V$, $\dim V = n$, B basis of V.

$\Rightarrow$ The map $\varphi : \mathrm{End}_K(V) \to \mathcal{M}_n(K)$, $\varphi(f) = [f]_B$, $(\forall) f \in \mathrm{End}_K(V)$ is an isomorphism of vector spaces and of rings.

<u>Proof</u>:

$(\mathrm{End}_K(V), +, \circ)$, $(\mathcal{M}_n(K), +, \cdot)$ rings.

? (The required isomorphisms follow by the above theor.

⑤ **Corollary**: Let $f \in End_K(V)$. Then $f \in Aut_K(V)$ $\Leftrightarrow \det([f]_B) \neq 0$

B any basis of V.

Proof:

$f \in Aut_K(V) \Leftrightarrow f$ invertible in $(End_K(V), +, \circ)$

$\Leftrightarrow [f]_B$ invertible in $(M_n(K), +, \cdot) \Leftrightarrow \det([f]_B) \neq 0$

Lecture 9

① **Theorem:** Let ${}_K V$. $B = (v_1, \dots, v_m)$
$B' = (v_1', \dots, v_m')$
$B'' = (v_1'', \dots, v_m'')$

$$\Rightarrow T_{BB''} = T_{BB'} \cdot T_{B'B''}$$

Proof:

$[f \circ g] = [f] \cdot [g]$

$$T_{BB'} \cdot T_{B'B''} = [1_V]_{B'B} \cdot [1_V]_{B''B'}$$
$$= [1_V \circ 1_V]_{B''B}$$
$$= [1_V]_{B''B}$$
$$= T_{BB''}.$$

② **Theorem:** ${}_K V$, B, B' bases of V.

$\Rightarrow T_{BB'}$ invertible with its inverse $= T_{B'B}$.

Proof:

Let $T_{B'B}$ be the inverse of $T_{BB'}$.

- $T_{BB'} \cdot T_{B'B} \xlongequal{\text{th. ①}} T_{BB} = I_m$.

(transition matrix from B to B) $\to$ identity

- $T_{B'B} \cdot T_{BB'} \xlongequal{\text{th. ①}} T_{B'B'} = I_m$

$$\Rightarrow T_{BB'}^{-1} = T_{B'B}$$

③ **Theorem**: ${}_K V$, $B = (v_1, \dots, v_m)$, $B' = (v_1', \dots, v_m')$ bases of V and $v \in V$. $\Rightarrow$ $\boxed{[v]_B = T_{BB'} \cdot [v]_{B'}}$.

Proof:

$$\underline{[f(v)]_{BB'} = [f]_{BB'} \cdot [v]_{B'}}$$

$$\underbrace{T_{BB'}} \cdot [v]_{B'} = \underbrace{[1_V]_{B'B}} \cdot [v]_{B'}$$

$$= [1_V(v)]_B$$

$$= [v]_B.$$

$$[v]_{B'} = T^{-1}_{BB'} \cdot [v]_B$$

$$= T_{B'B} \cdot [v]_B$$

④ **Theorem**: $f \in \mathrm{Hom}_K(\cancel{K}V, V')$

B_1, B_2 bases of V

B_1', B_2' bases of V'.

$$\Rightarrow [f]_{B_2B_2'} = T^{-1}_{B_1'B_2'} \cdot [f]_{B_1B_1'} \cdot T_{B_1B_2}$$

Proof:

$$T^{-1}_{B_1'B_2'} \cdot [f]_{B_1B_1'} \cdot T_{B_1B_2} =$$

$$= T_{B_2'B_1'} \cdot [f]_{B_1B_1'} \cdot T_{B_1B_2}$$

$$= [1_{V'}]_{B_1'B_2'} \cdot [f]_{B_1B_1'} \cdot [1_V]_{B_2B_1}$$

$$\underline{[f \circ g] = [f] \cdot [g]}$$

$$= [1_{V'} \circ f \circ 1_V]_{B_2B_2'}$$

$$= [f]_{B_2B_2'}$$

⑤. **Theorem**: $V(\lambda) = \{v \in V \mid f(v) = \lambda v\}$

is a subspace of V; $f \in \mathrm{End}_K(V)$.

Proof:

$0 \in V(\lambda) \Rightarrow V(\lambda) \neq \emptyset$ (1)

$k_1, k_2 \in K$

$v_1, v_2 \in V(\lambda)$

$f(v_1) = \lambda v_1$

$f(v_2) = \lambda v_2$

$f(k_1 v_1 + k_2 v_2) \overset{\text{lin. map.}}{=\!=\!=} k_1 \underline{f(v_1)} + k_2 \underline{f(v_2)}$

$= k_1 (\lambda v_1) + k_2 (\lambda v_2)$

$= (k_1 \lambda) v_1 + (k_2 \lambda) v_2$

$= \lambda \underbrace{(k_1 v_1 + k_2 v_2)}_{\in V(\lambda)} \Rightarrow V(\lambda)$ subspace of V

?

⑥ **Theorem**: ${}_K V$, B basis of V

$f \in \mathrm{End}_K(V)$ with $[f]_B = A = (a_{ij}) \in \mathcal{M}_m(K)$.

$\Rightarrow \lambda \in K$ eigenvalue of $f \Leftrightarrow \det(A - \lambda I_m) = 0$

Proof:

$\lambda \in K$ eigenvalue of $f \Leftrightarrow \exists v \neq 0 \in V$ s.t.

$f(v) = \lambda v$

$$[v]_B = \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix}$$

$f(v) = \lambda v$

$\Leftrightarrow f(v) - \lambda v = 0$

$\Leftrightarrow (f - \lambda \cdot 1_V)(v) = 0$

$\xleftarrow{\text{v}\to\text{matrix}} [(f-\lambda\cdot 1_V)(v)]_B = [0]_B$

$\Leftrightarrow [f - \lambda \cdot 1_V]_B \cdot [v]_B = [0]_B$

$\Leftrightarrow ([f]_B - \lambda [1_V]_B) \cdot [v]_B = [0]_B$

$\Leftrightarrow (A - \lambda \cdot I_m) \cdot [v]_B = [0]_B$

$$\Leftrightarrow \begin{pmatrix} a_{11}-\lambda & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22}-\lambda & \dots & a_{2m} \\ \vdots & \vdots & & \vdots \\ a_{m1} & \dots & a_{m2} \dots & a_{mm}-\lambda \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_m \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$\Leftrightarrow (S) \begin{cases} (a_{11}-\lambda)x_1 + a_{12}\cdot x_2 + \dots + a_{1m}\cdot x_m = 0 \\ a_{21}x_1 + (a_{22}-\lambda)x_2 + \dots + a_{2m}\cdot x_m = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + (a_{mm}-\lambda)x_m = 0 \end{cases}$$

$\Rightarrow \lambda$ eigenvalue of $f \Leftrightarrow (S)$ has a non-zero solution $\Leftrightarrow \underbrace{\det(A - \lambda I_m) = 0}_{\text{CHARACTERISTIC EQUATION}}$.

(S) CHARACTERISTIC SYSTEM; $\underbrace{P_A(\lambda) = \det(A - \lambda I_m)}_{\text{CHARACTERISTIC POLYNOMIAL}}$

(7) **Theorem:** ${}_K V$, B, B' bases of V.

$f \in End_K(V)$, $[f]_B = A \in \mathcal{M}_m(K)$

$[f]_{B'} = A' \in \mathcal{M}_m(K)$

$\Rightarrow \boxed{P_A(\lambda) = P_{A'}(\lambda)}$

Proof:

$[f]_{B'} = T^{-1}_{BB'} \cdot [f]_B \cdot T_{BB'}$

$A' = T^{-1} \cdot A \cdot T$

$$
\begin{aligned}
P_{A'}(\lambda) &= \det(A' - \lambda I_m) \\
&= \det(T^{-1} \cdot A \cdot T - \lambda I_m \cdot T^{-1} \cdot T) \\
&= \det(T^{-1}(A - \lambda I_m)T) \\
&= \det(T^{-1}) \cdot \det(A - \lambda I_m) \cdot \det(T) \\
&= \det(A - \lambda I_m) \\
&= P_A(\lambda)
\end{aligned}
$$